13. Spotted lanternfly (*Lycorma delicatula*)

Last modified: October 16, 2026

13A. Introduction

The spotted lanternfly (SLF), *Lycorma delicatula* (White) (Hemiptera: Fulgoridae), is a highly polyphagous planthopper that feeds on a broad range of agricultural and non-agricultural plants, including grapevines, fruit and nut trees, and ornamental species (Chu 1930, Barringer and Ciafré 2020, EPPO 2025). Adults cause direct damage by feeding on plant sap, while the accumulation of honeydew promotes the growth of sooty mold, which can inhibit photosynthesis and attract additional pests (Dara et al. 2015, Urban and Leach 2023). Native to China, SLF was first detected in the United States in 2014 in Pennsylvania and has since spread to at least 20 eastern states (Urban and Leach 2023, NYSIPM 2025). The widespread presence of its preferred host, tree-of-heaven [*Ailanthus altissima* (Mill.) Swingle], has facilitated its rapid expansion (Barringer and Ciafré 2020). In suburban and urban environments, SLF can become a significant nuisance, with large aggregations of adults often swarming on trees (Urban and Leach 2023). The species has already caused substantial economic impacts, including estimated annual losses of $8.8 million to New York's grape industry alone (Pinto et al. 2025).

Modeling studies indicate that SLF has not yet filled its fundamental niche in the contiguous United States (CONUS) under current weather patterns, with suitable conditions extending across much of the Pacific Coast and eastern United States (Jones et al. 2022, Wakie et al. 2019, Barker et al. 2025). The coldest regions of the CONUS, particularly portions of the Midwest and Northeast, may be less suitable for establishment (Wakie et al. 2019, Barker et al. 2025), although urban heat island effects could facilitate establishment in colder areas beyond the currently predicted range (Owen et al. 2024). SLF phenology may also shift in response to changing weather patterns. Analysis of citizen science observations from 2014 to 2022 found earlier adult emergence and later winter die-off as SLF expanded its range in the eastern United States (Owen et al. 2024). However, the effects of temperature, photoperiod, and other environmental factors on egg diapause and developmental rates remain incompletely understood, limiting predictions of how SLF may respond to changing weather patterns (Keena and Nielsen 2021, Calvin et al. 2023). Diapause may buffer populations against weather variability by enhancing overwintering egg survival and accelerating development and fecundity during summer and fall heat waves (Lewkiewicz et al. 2024). SLF is currently considered univoltine, including in warmer regions of China, and whether it can complete multiple generations annually in warmer areas of the United States remains uncertain (Keena and Nielsen 2021, Calvin et al. 2023).

A model for spotted lanternfly developed for the Degree-Day, establishment Risk, and Phenological event mapping system (DDRP) is operationalized to produce near real-time forecasts for the CONUS at https://uspest.org/DDRP. The phenology model applies a lower developmental threshold of 10 °C and upper threshold of 35 °C for all life stages (egg, larva, pupa, and adult) and assumes the egg is the overwintering stage. The climatic suitability model applies a lower cold stress threshold and upper heat stress threshold of −16 °C and 37 °C, respectively. The full set of model parameters for spotted lanternfly, including degree-day requirements for individual life stages and climate stress limits, are presented in Table 1. Details on model parameterization are available at USPest.org (Barker et al. 2025).

Table 1. Spotted lanternfly model parameters for DDRP.

Parameter	Code	Value
Lower developmental thresholds (°C)		

Parameter	Code	Value
Egg	eggLDT	10
Larvae	larvaeLDT	10
Pupae	pupaeLDT	10
Adult	adultLDT	10
Upper developmental thresholds (°C)		
Egg	eggUDT	35
Larvae	larvaeUDT	35
Pupae	pupaeUDT	35
Adult	adultUDT	35
Stage durations (°C degree-days)		
Egg	eggDD	202
Larvae	larvaeDD	890
Pupae	pupDD	630
Adult	adultDD	146
Pest events (°C degree-days)		
J-larval development complete	OWEventDD	202
Egg hatch	eggEventDD	100
Developing J-larvae	larvaeEventDD	442
Adult emergence (pupal development complete)	pupaeEventDD	1
Egg laying	adultEventDD	1
Cold stress		
Cold stress temperature threshold (°C)	coldstress_threshold	-16
Cold degree-day (°C) limit when most individuals die	coldstress_units_max1	300
Cold degree-day (°C) limit when all individuals die	coldstress_units_max2	475
Heat stress		
Heat stress temperature threshold (°C)	heatstress_threshold	37
Heat stress degree-day (°C) limit when most individuals die	heatstress_units_max1	115
Heat stress degree-day (°C) limit when all individuals die	heatstress_units_max2	175
Cohorts		
Degree-days (°C) to emergence (average)	distro_mean	190
Degree-days (°C) to emergence (variation)	distro_var	15000
Minimum degree-days (°C) to emergence	xdist1	135
Maximum degree-days (°C) to emergence	xdist2	360
Shape of the distribution	distro_shape	normal
Other		
Order of stages	stgorder	OE, L, P, A, E
Degree-day calculation method	calctype	triangle

13B. Results

i. Phenology

Entire period: 1981–2025

Over 1981–2025, predicted first egg hatch and adult emergence showed mixed spatial trends, with later events occurring across portions of the northern and central United States and earlier events in scattered southern and western areas (Figure 1, Figure 2). Significant changes were concentrated in the eastern and central United

States and in localized western areas.

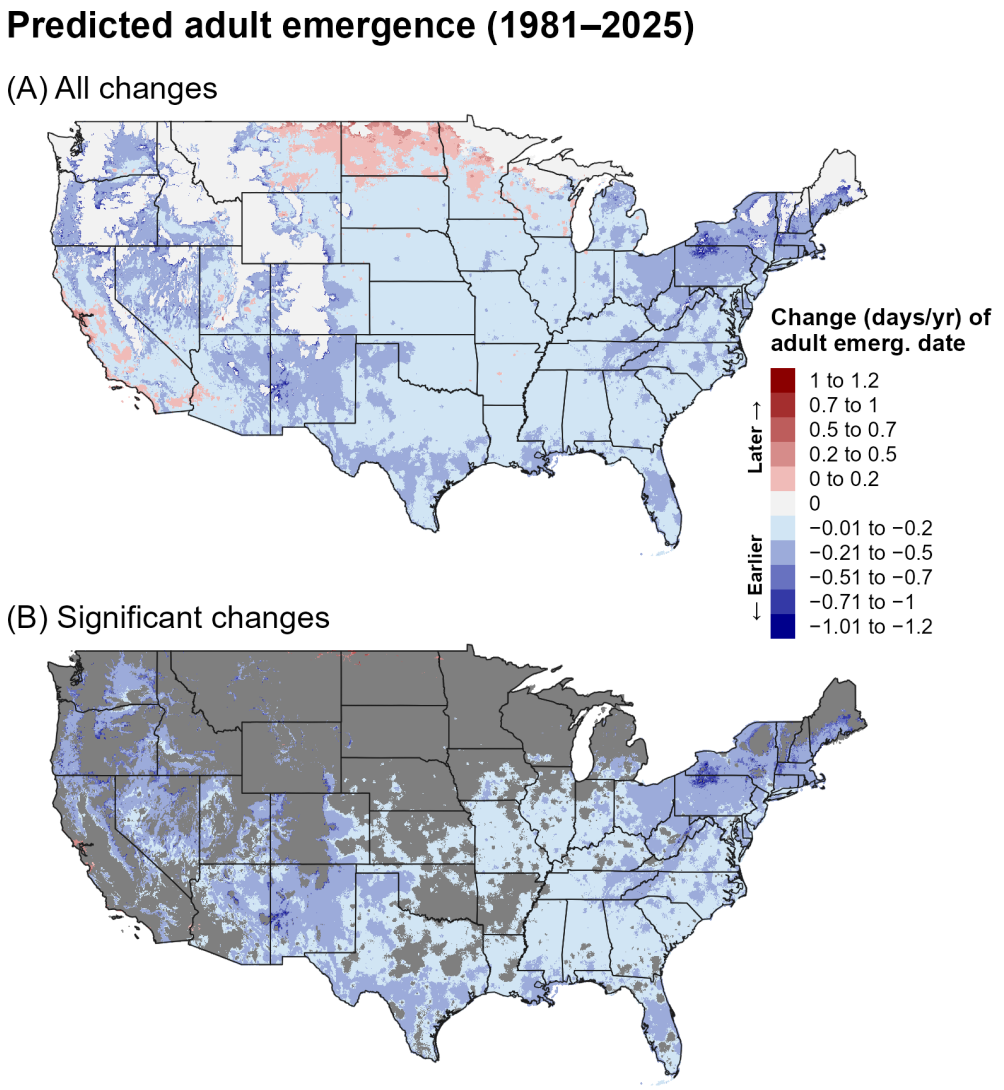


Figure 1. Changes (days/year) in predicted dates of first adult emergence for Spotted lanternfly across the CONUS over the entire 44-year period (1981–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant.

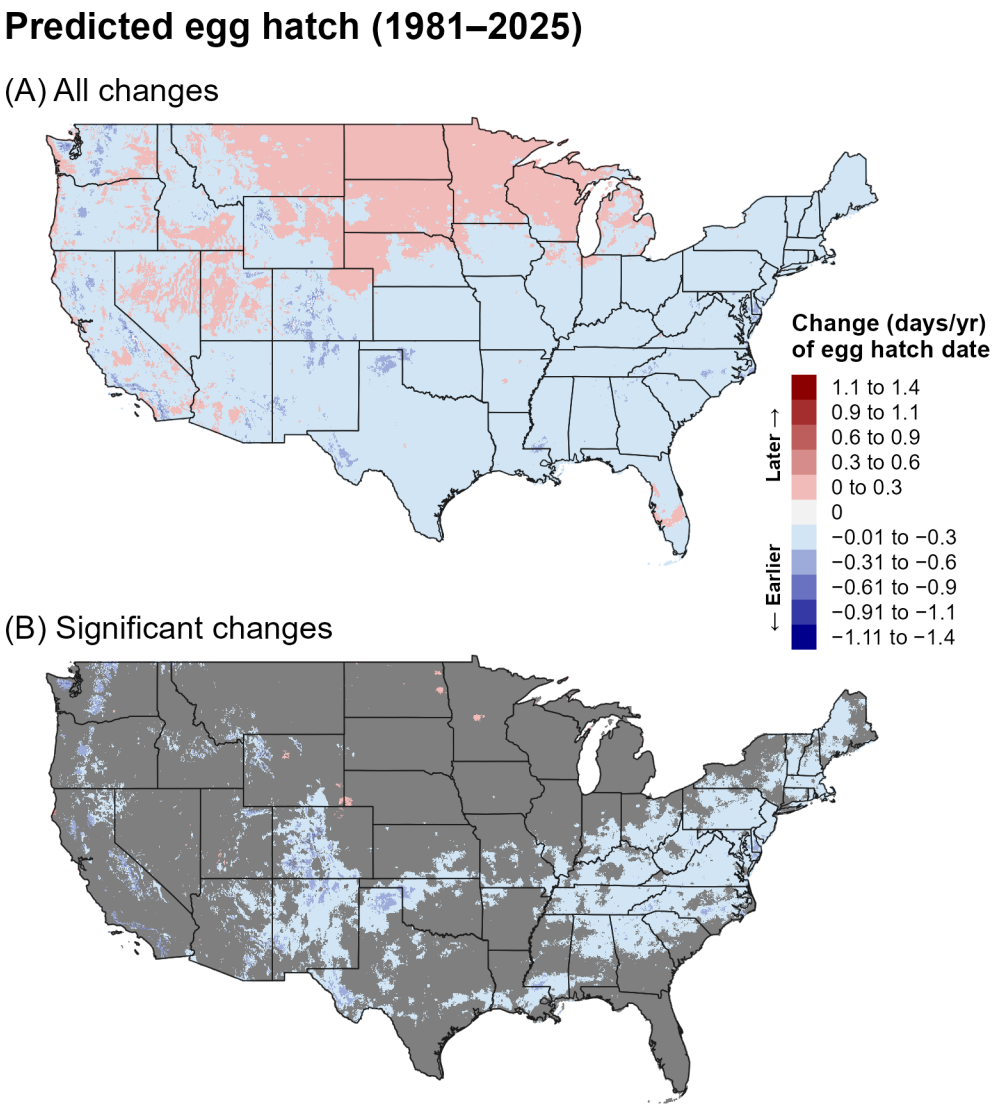


Figure 2. Changes (days/year) in predicted dates of first egg hatch for Spotted lanternfly across the CONUS over the entire 44-year period (1981–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant.

Recent period: 2001–2025

During 2001–2025, later predicted egg hatch and adult emergence were more apparent across portions of the northern and western United States, while earlier events occurred in scattered central and eastern areas (Figure 3, Figure 4). Significant changes were relatively limited in spatial extent.

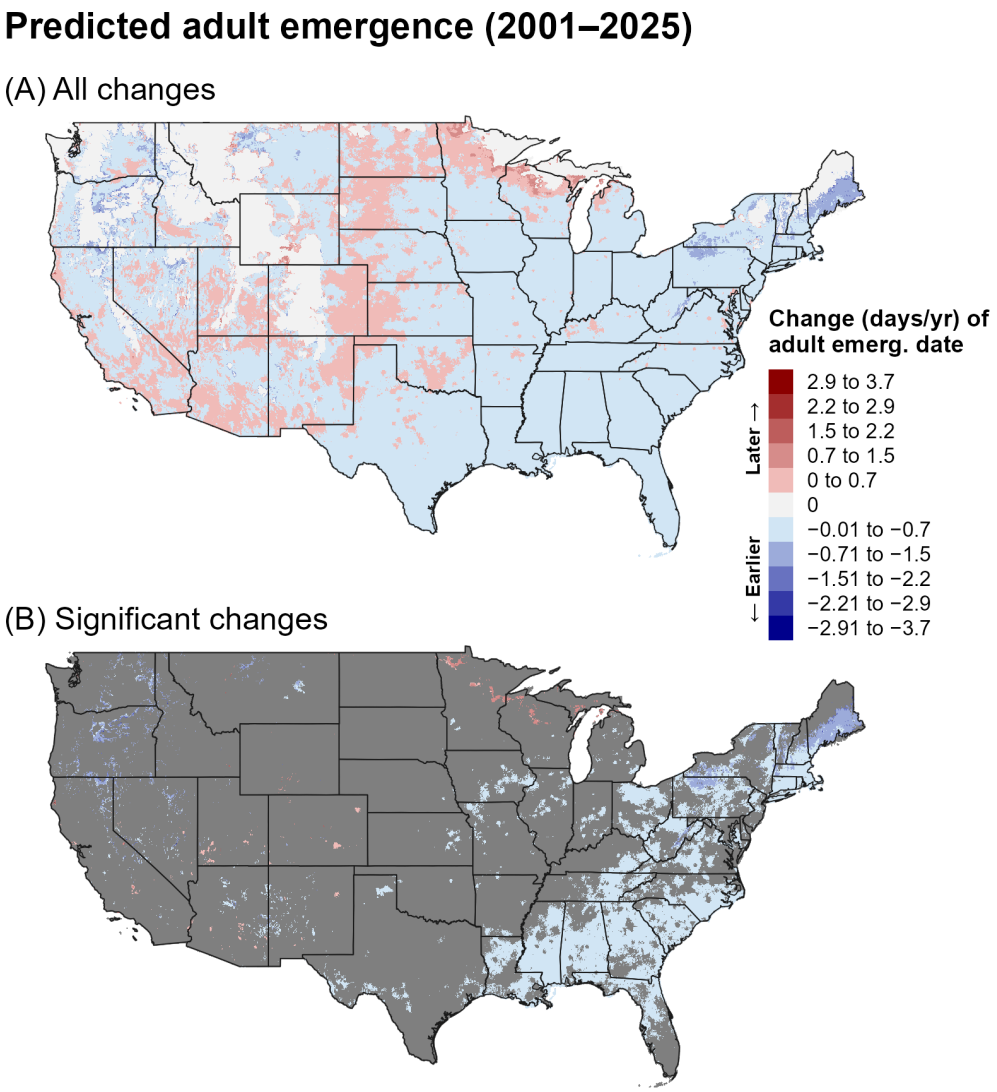


Figure 3. Changes (days/year) in predicted dates of first adult emergence for Spotted lanternfly across the CONUS over the recent 24-year period (2001–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant.

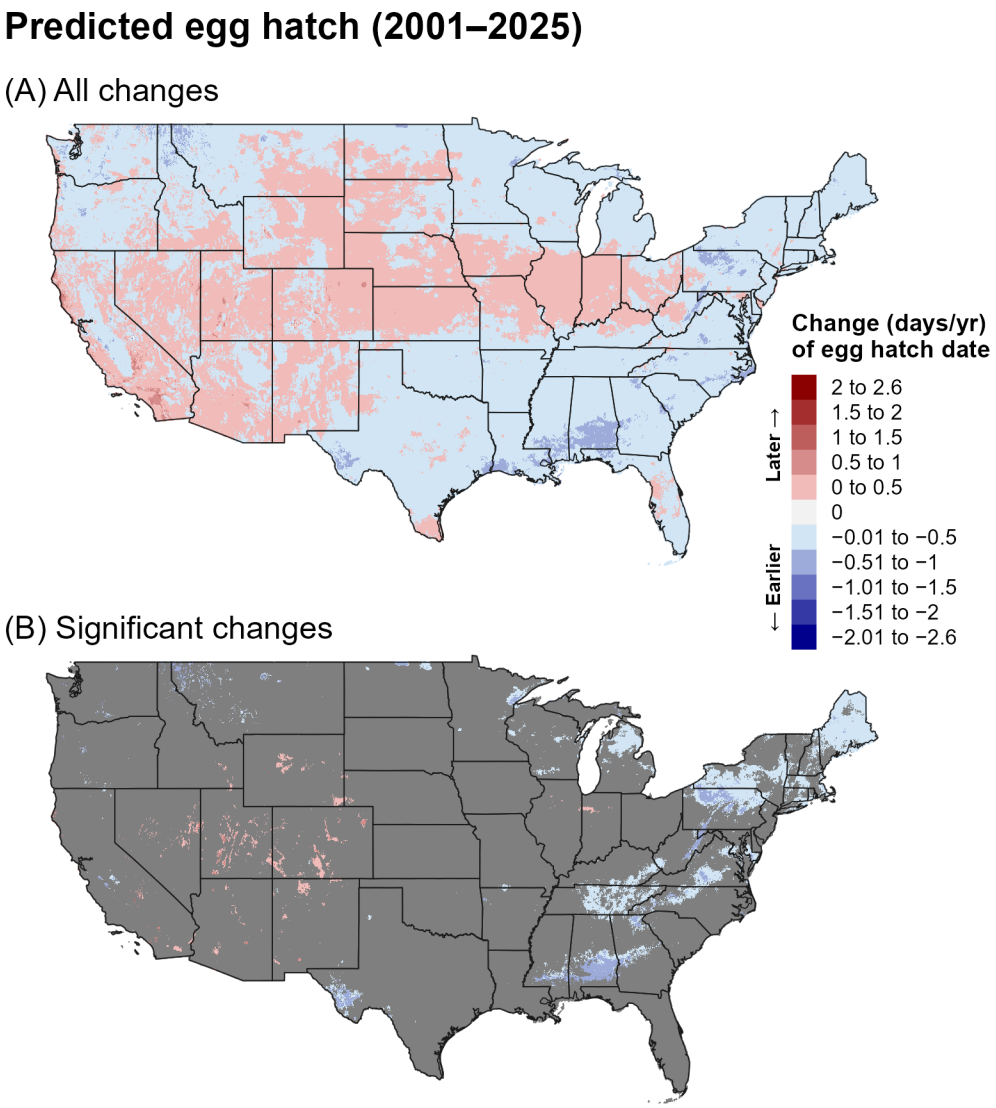


Figure 4. Changes (days/year) in predicted dates of first egg hatch for Spotted lanternfly across the CONUS over the recent 24-year period (2001–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant.

ii. Climatic suitability

Entire period: 1981–2025

Predicted cold stress decreased across much of the northern United States over 1981–2025, whereas heat stress increased in the western and southern United States (Figure 5, Figure 6). Climatic-suitability changes were limited, with significant increases occurring mainly in localized northern areas (Figure 7).

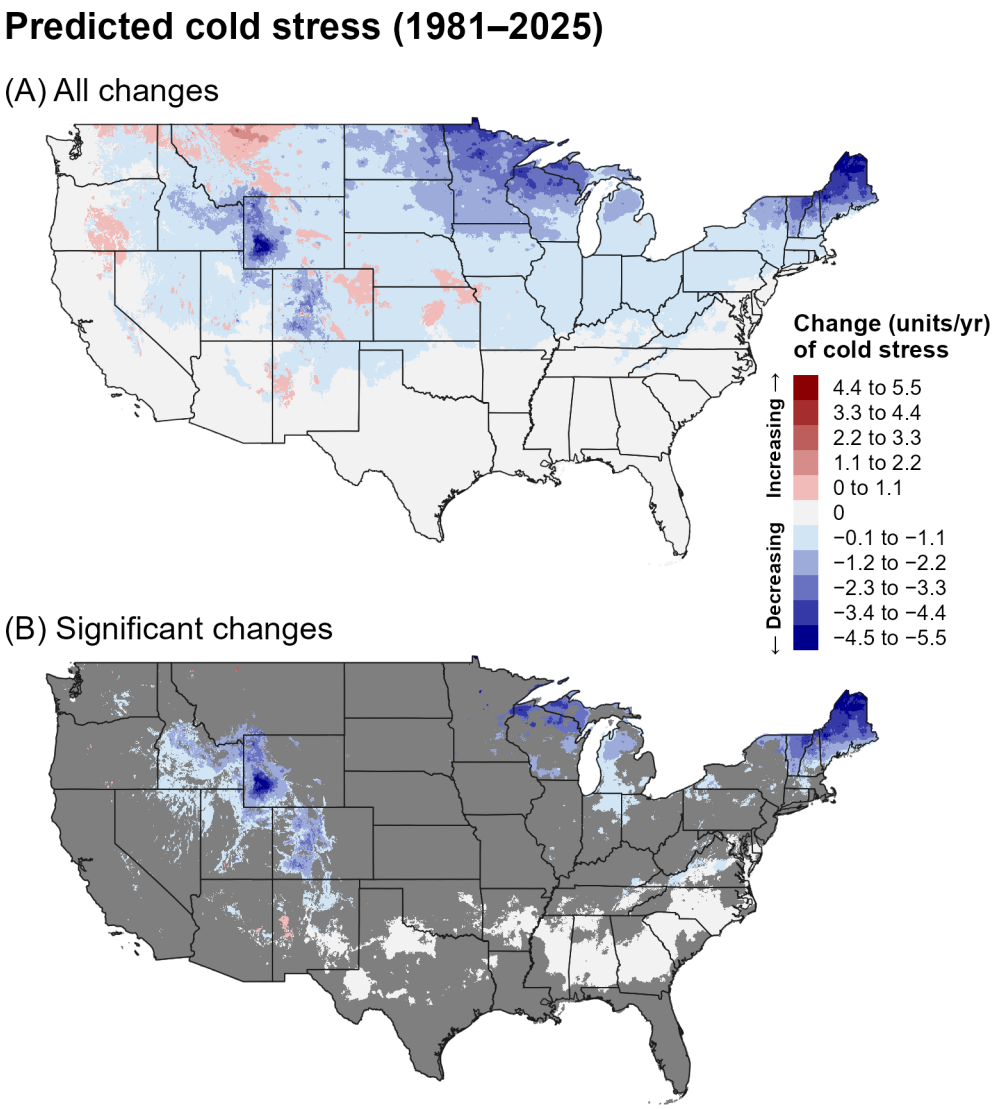


Figure 5. Changes (units/year) in predicted cold stress for Spotted lanternfly across the CONUS over the entire 44-year period (1981–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant. Areas shown as 0 in (B) had significant trends but very small changes.

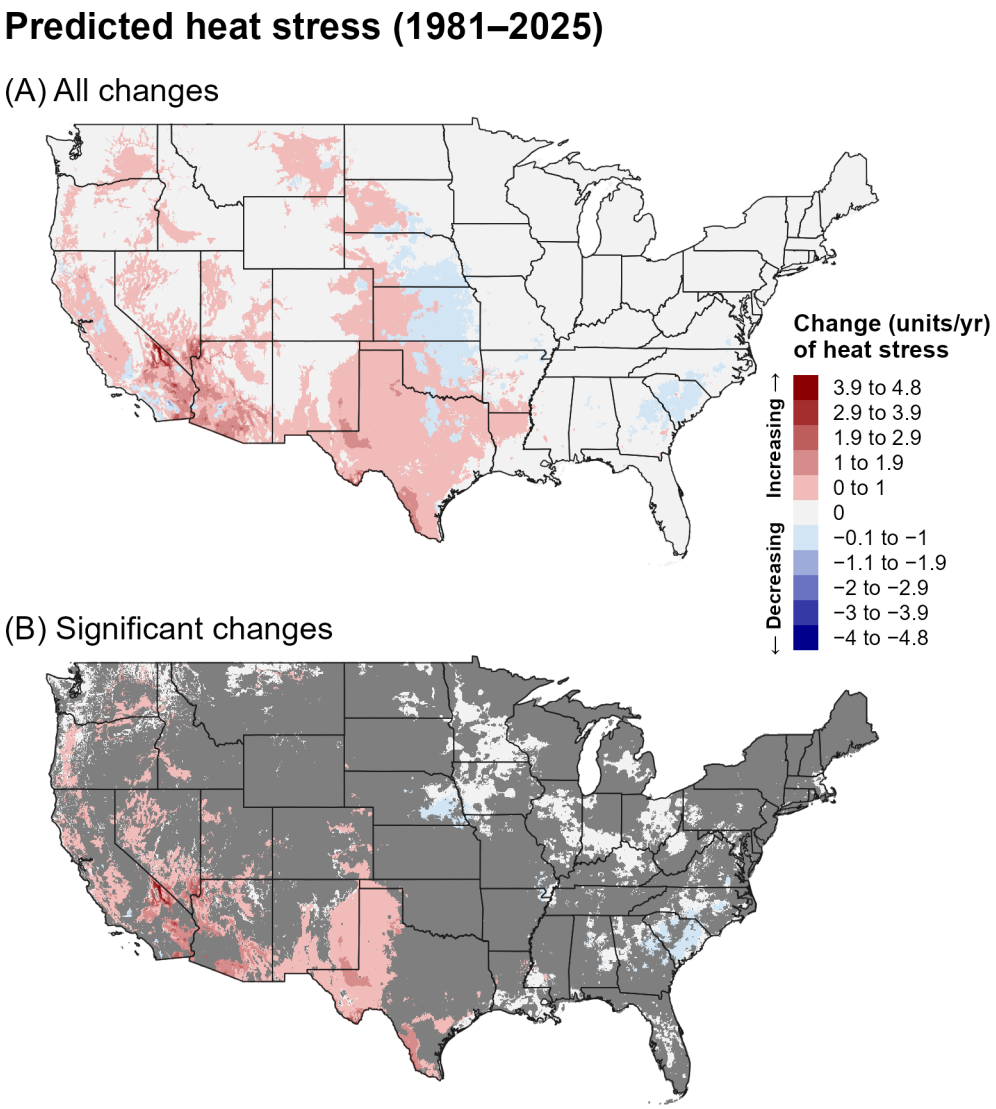


Figure 6. Changes (units/year) in predicted heat stress for Spotted lanternfly across the CONUS over the entire 44-year period (1981–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant. Areas shown as 0 in (B) had significant trends but very small changes.

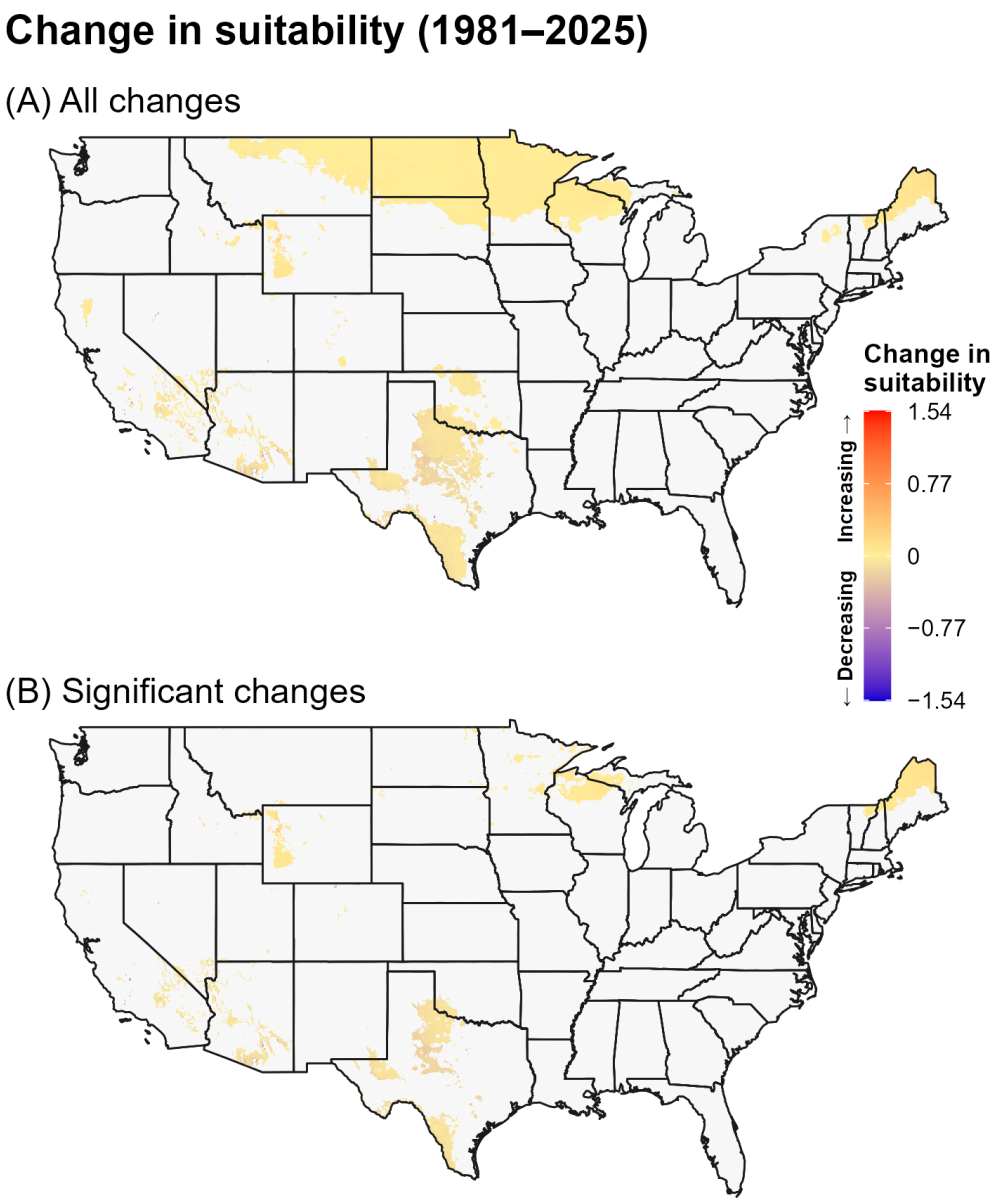


Figure 7. Changes in predicted climatic suitability for Spotted lanternfly across the CONUS over the 44-year period (1981–2025). Suitability is defined as areas not excluded by climate stress. Positive changes (warmer colors) indicate an increased likelihood of establishment, whereas negative changes (cooler colors) indicate a decreased likelihood of establishment.

Recent period: 2001–2025

During 2001–2025, increases in cold stress became more evident across portions of the northern and central interior, while heat stress continued to increase in the Southwest and southern Great Plains (Figure 8, Figure 9). Significant suitability changes remained highly localized (Figure 10).

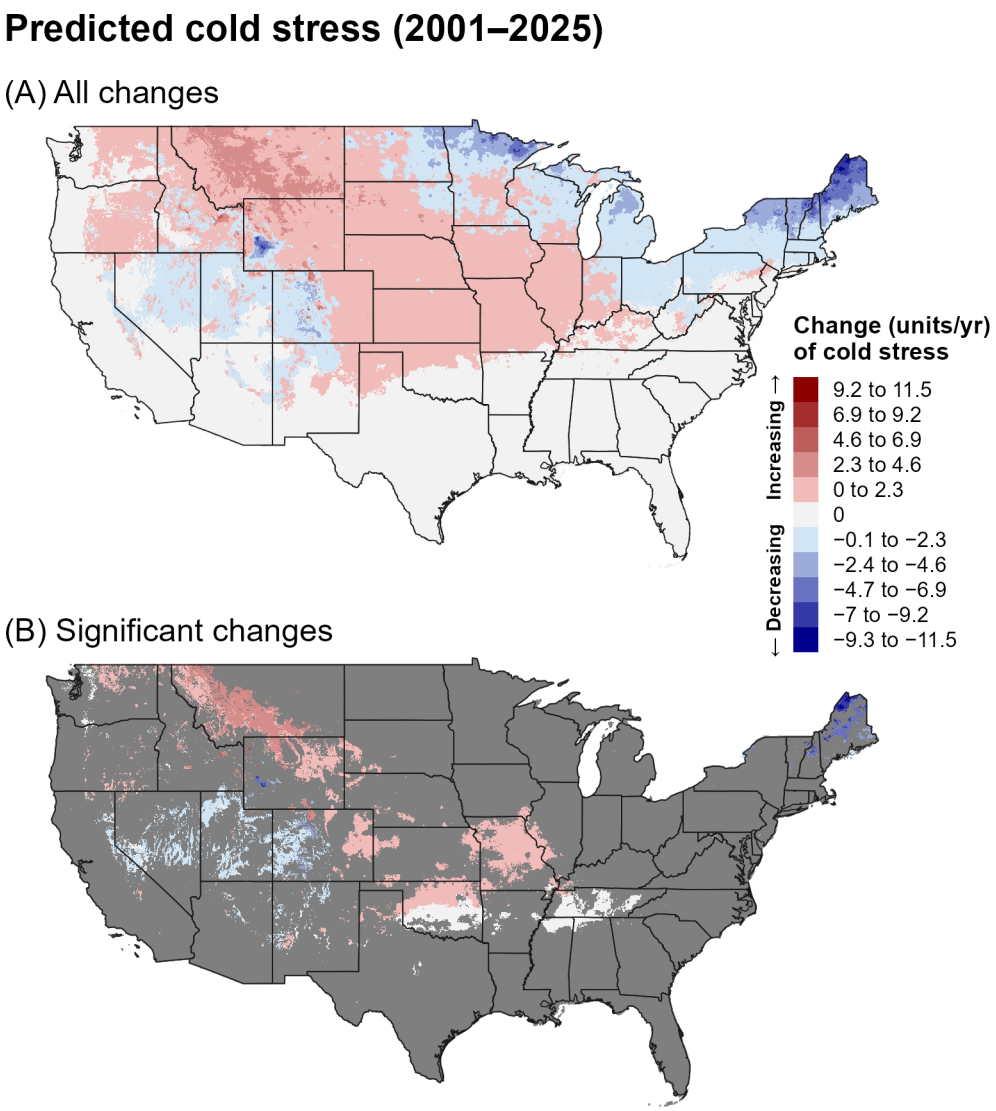


Figure 8. Changes (units/year) in predicted cold stress for Spotted lanternfly across the CONUS over the recent 24-year period (2001–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant. Areas shown as 0 in (B) had significant trends but very small changes.

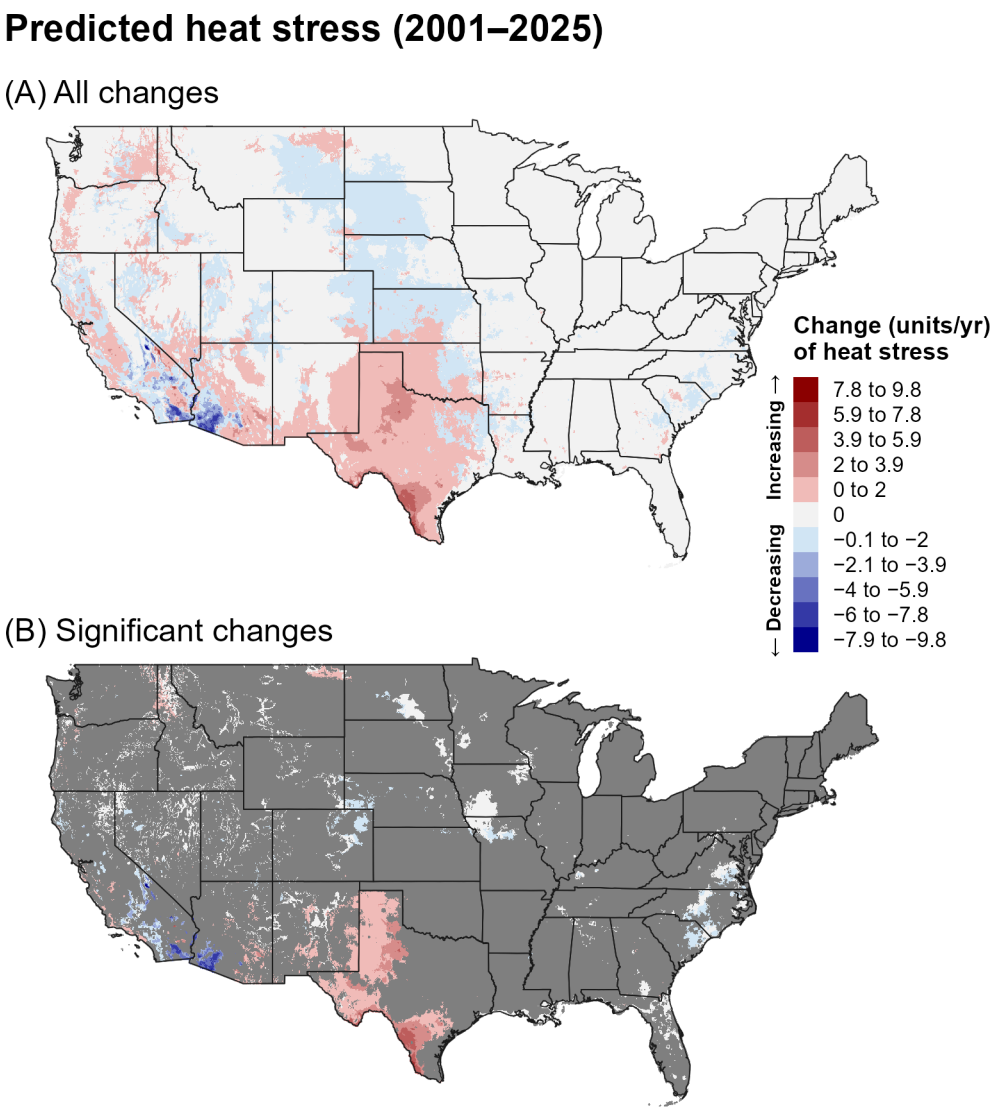


Figure 9. Changes (units/year) in predicted heat stress for Spotted lanternfly across the CONUS over the recent 24-year period (2001–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant. Areas shown as 0 in (B) had significant trends but very small changes.

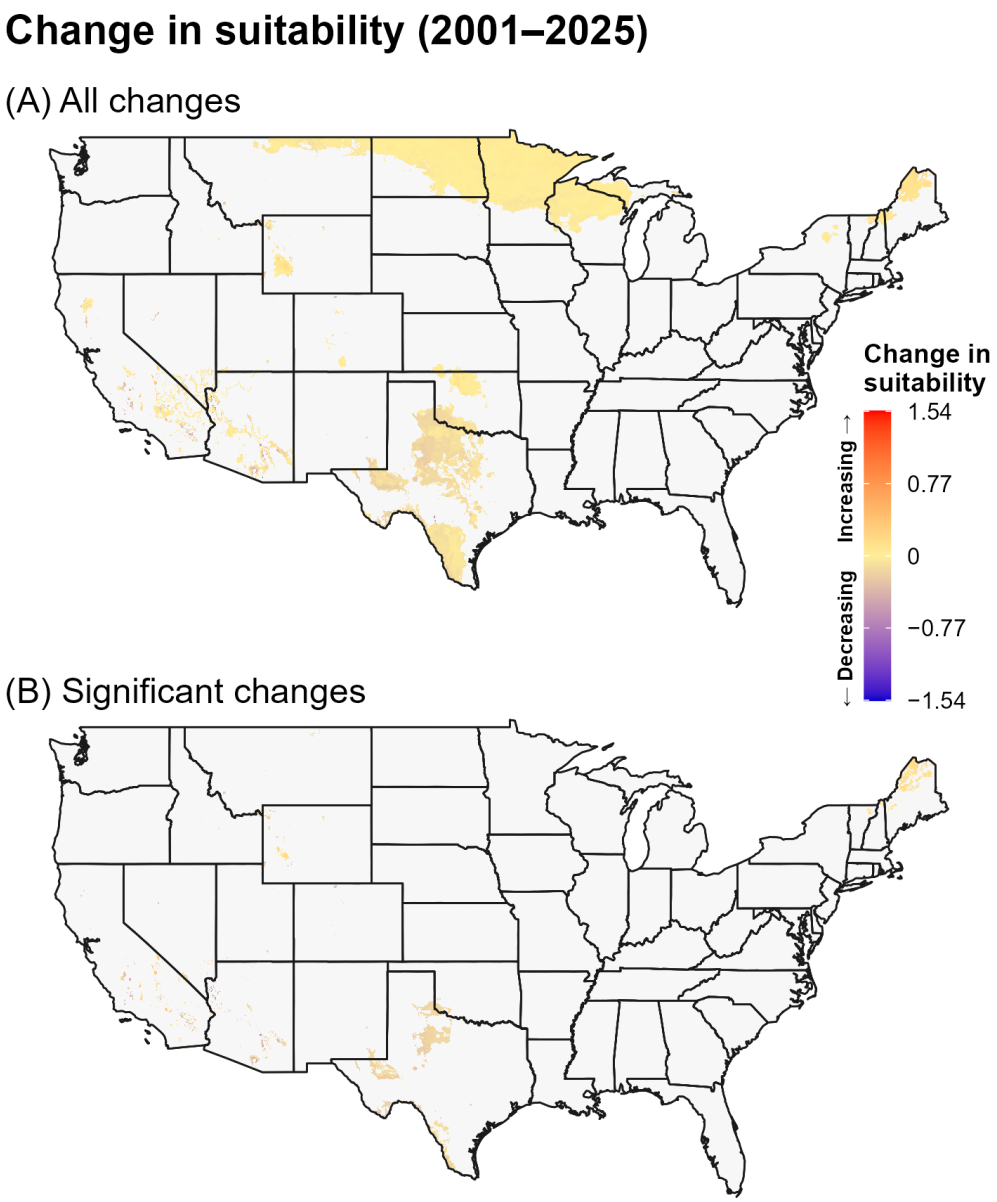


Figure 10. Changes in predicted climatic suitability for Spotted lanternfly across the CONUS over the recent 24-year period (2001–2025). Suitability is defined as areas not excluded by climate stress. Positive changes (warmer colors) indicate an increased likelihood of establishment, whereas negative changes (cooler colors) indicate a decreased likelihood of establishment.

13C. Discussion

The SLF maps show a mixed phenological response, with the direction of change varying geographically and between the two periods. The absence of a single dominant spatial direction suggests that the timing of SLF development may respond differently to regional climate trends. Because the species is already capable of substantial spread in North America, localized changes in predicted phenology may be particularly relevant for surveillance timing in newly suitable areas.

Although cold and heat stress changed across broad areas, relatively little of that variation translated into significant changes in predicted climatic suitability. The long-term reduction in cold stress across northern

regions could reduce one climatic constraint on establishment, but recent increases in cold stress in portions of the northern interior may have moderated this trend.

The stronger and more spatially extensive trends in predicted phenology and climate stress over the full 44-year period may reflect both the larger sample size, which can reduce noise and improve detection of directional patterns, and differences in the underlying climate trends between the two time periods. In particular, the recent 24-year period may capture climate variability or regional trends that differ from those observed over the longer 44-year period. Thus, differences between the two periods should be interpreted as reflecting both long-term climate trends and shorter-term regional variability.

13D. References

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